Seok Jung Kim

M.S. Student, Dept. of Industrial and Management Systems Engineering, Kyung Hee University
ff9815@khu.ac.kr | sierrajulliet97.github.io

Research Interests

- Drone vehicle routing and last-mile delivery under urban wind
- Hybrid metaheuristics (VNS, ALNS) for large-scale routing problems
- Learning-assisted optimization (reinforcement learning within metaheuristics)

Education

Kyung Hee University Sep. 2026 – Present
M.S. in Industrial and Management Systems Engineering (IO-LAB) Yongin, Korea

Chonnam National University Mar. 2023 – Aug. 2025
B.B.A. in Business Administration / Double Major in Big Data Convergence Gwangju, Korea

Research Experience

Wind- and Level-aware Drone Vehicle Routing Problem (WiLD-VRP) Sep. 2026 – Present
Advisor: Prof. Ho Young Jeong

- Formulate a multi-trip drone routing problem in which the flight level of each route segment is chosen jointly with the visiting sequence under altitude-dependent urban wind.
- Develop a variable neighborhood search (VNS) with an offline Q-learning rule that skips unpromising move evaluations; benchmark against SA, ALNS, and exact MILP solutions (Gurobi).
- Build test instances from real building data for five districts of Seoul.
- Manuscript in preparation for *Transportation Research Part E*.

Presentations

S. J. Kim and H. Y. Jeong, "Segment-wise altitude selection in urban drone delivery routing under wind," Asia Pacific Industrial Engineering & Management Systems Conference (APIEMS 2026), Busan, Republic of Korea, Dec. 2026 (scheduled, oral).

Honors & Awards

Encouragement Award, The 4th Joint Academic Seminar, Kyung Hee University Nov. 2026
For a presentation of the WiLD-VRP research

Activities and Experience

KDI Knowledge Sharing Program (KSP) 2024 – 2025
Young KSP (YKSP) Cohort 8 – Infrastructure & Transportation Information Systems, Honduras; two field dispatches (2024, 2025)

Korea–Arab Youth Exchange Program 2025
Sponsored by S-OIL; field exchange to the UAE (Dubai) and Saudi Arabia

LG Aimers Cohort 6 2024
AI/ML Education Program, LG AI Research

Certifications & Skills

- **Programming & Tools**: Python, C++, Gurobi
- **Methods**: Mixed-integer linear programming (MILP), variable neighborhood search (VNS), adaptive large neighborhood search (ALNS), reinforcement learning (Q-learning)
- **Certification**: ADsP (Advanced Data Analytics Semi-Professional), 2024
- **Language**: TOEIC 910 (ETS), 2024
- **Volunteer**: World Friends Korea (WFK) online education volunteer for students in Madagascar.